Swiss Institute of Bioinformatics

PR STATS

# Metagenomic Data Analysis of Microbial Communities (MGMC01)

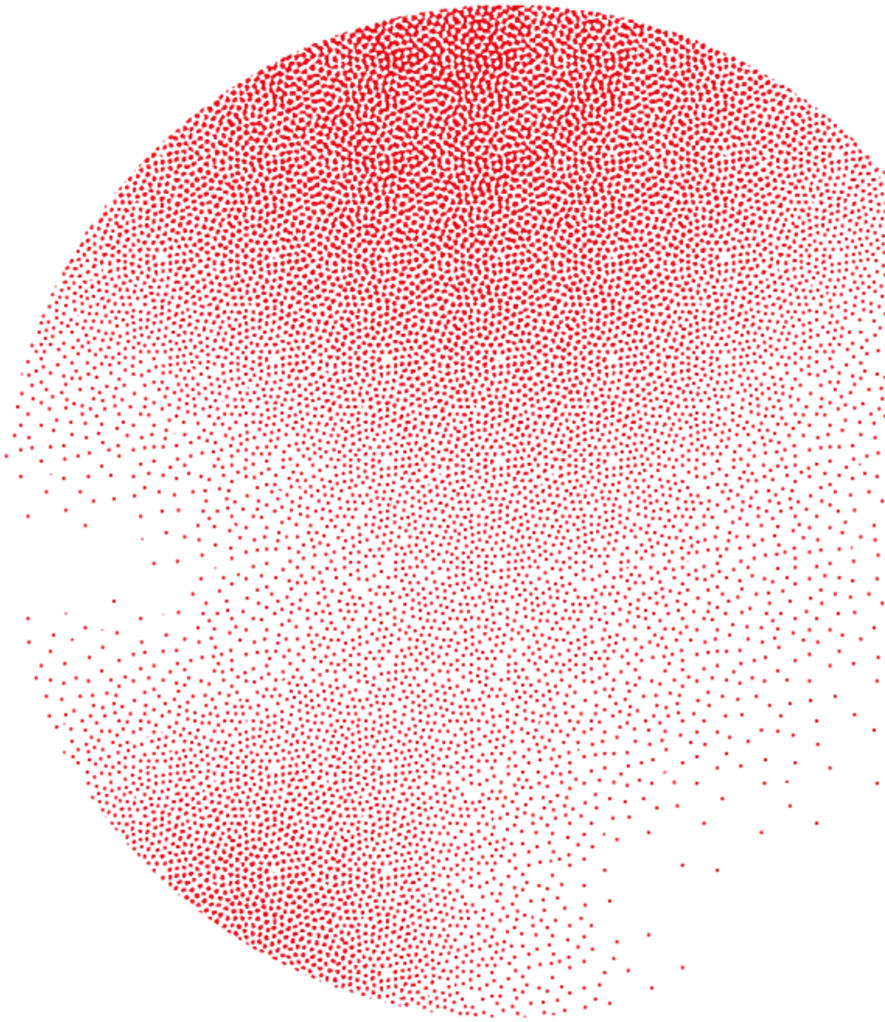
## Day 1 - Session 2

PhD Jeferyd Yepes-García
University of Fribourg
October 26th, 2026

elixir SWITZERLAND

# This presentation

- Learning outcomes
- Sequencing methods
- Sequencing by synthesis
- Long-read technologies
- Data cleaning and quality control
- Summary



# Learning outcomes

- Recognize the main sequencing technologies and how the choice of short- or long-read platforms influences the experimental design and the analysis strategy.
- Explain why quality control (QC) is a critical step in any metagenomics workflow.
- Interpret the main modules of a FastQC report and compare raw vs cleaned reads.

# Reminder of the sequencing methods




1986 - 2001 | 2005 | 2006 | 2010 | 2011 | 2014 | 2018

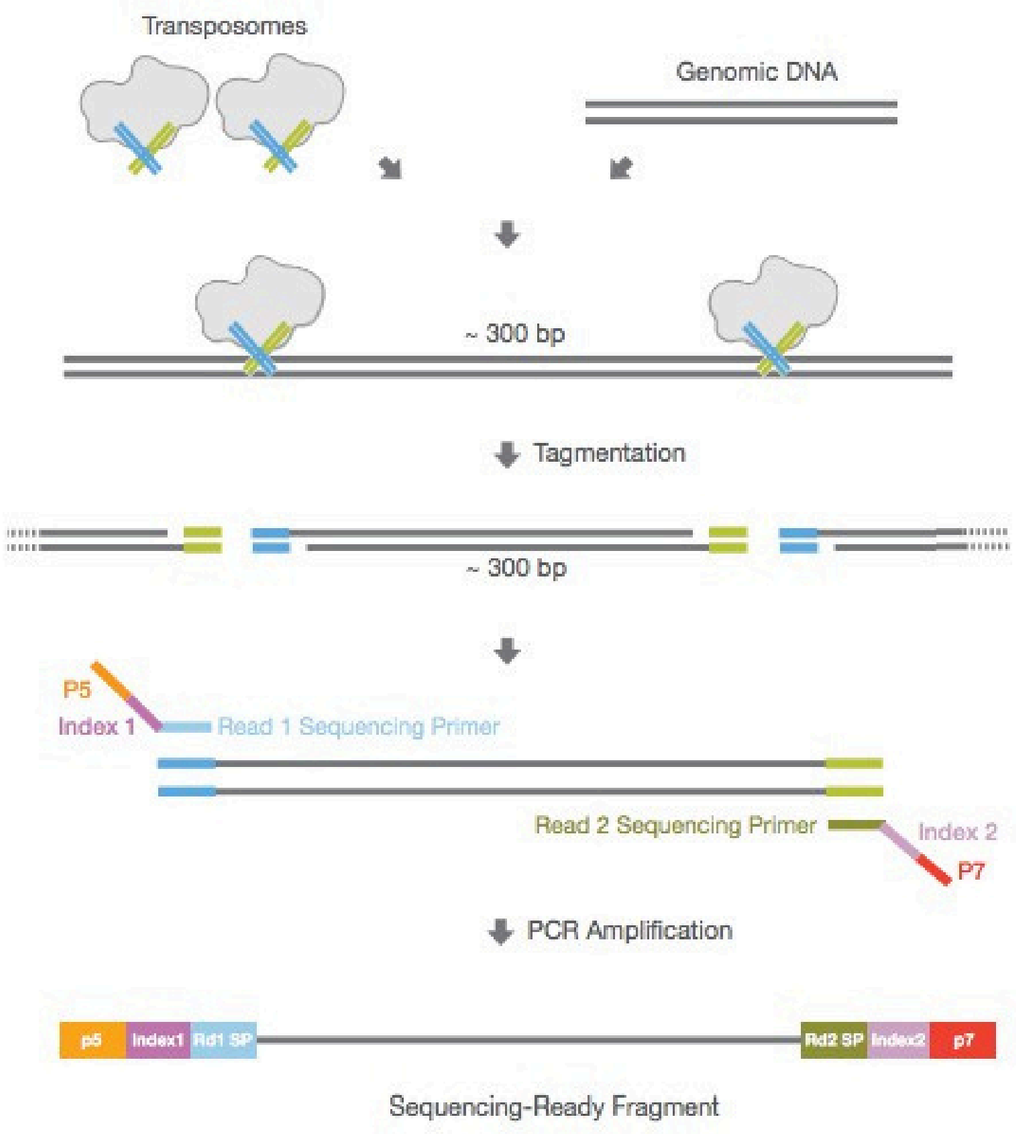
ABI Sanger | Roche 454 | Illumina | Life SOLiD | Life Ion Torrent | Pacific Biosciences | Oxford Nanopore

First Generation

Second Generation

Third Generation

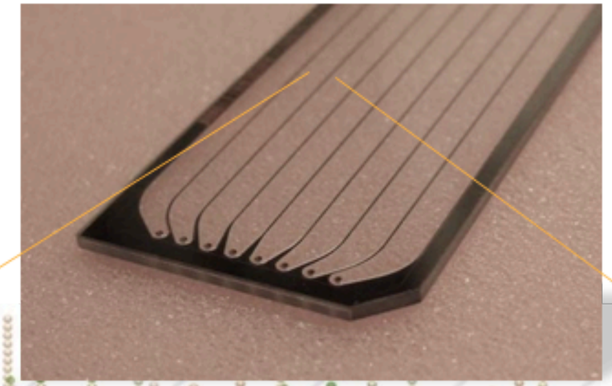
# The most used sequencing method: Sequencing by synthesis

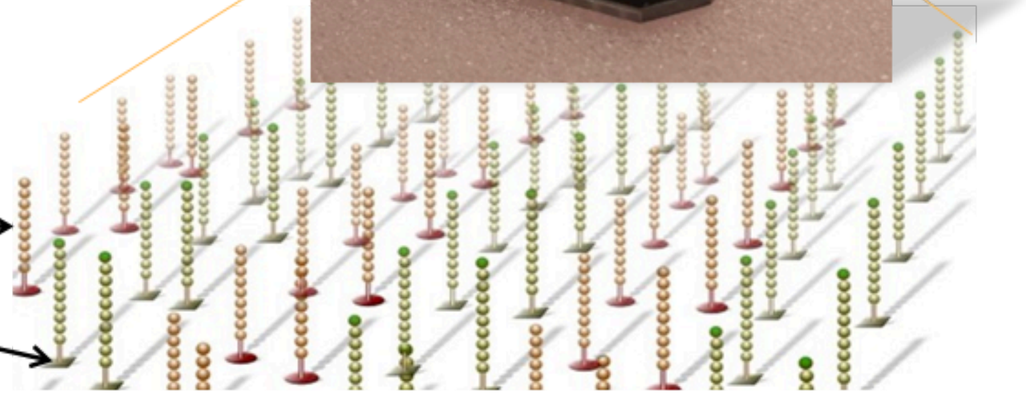


# The most used sequencing method: Sequencing by synthesis

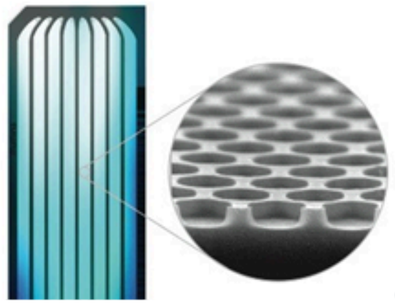
A flow cell is a thick glass slide with several channels or lanes

Each lane is randomly coated with a lawn of oligos that are complementary to library adapters or patterned



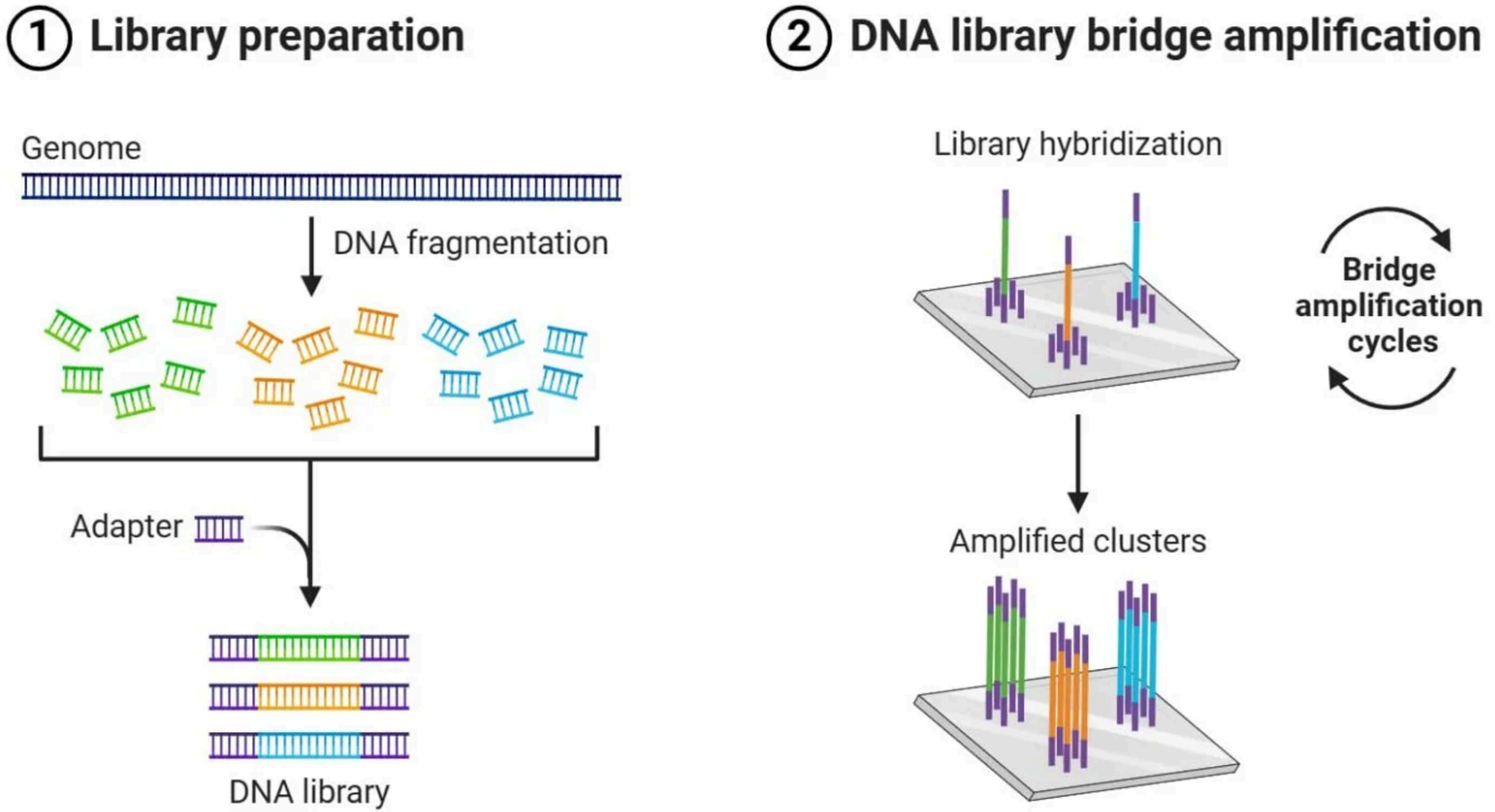




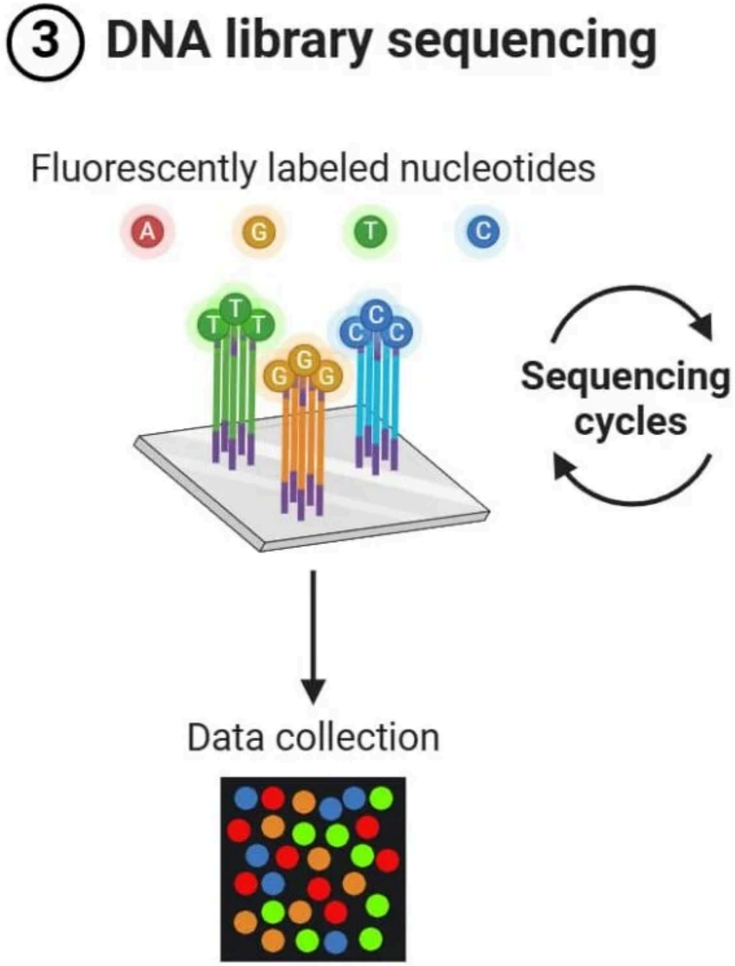
Patterned Flow Cells: higher number of clusters

# The most used sequencing method: Sequencing by synthesis




Tamang (2024)

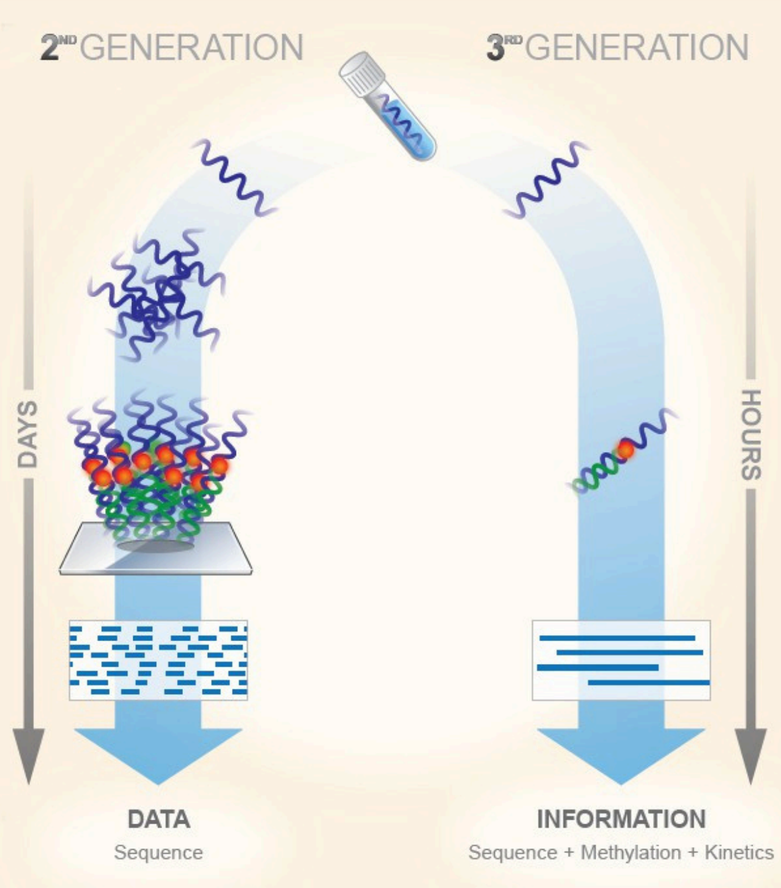
# The most used sequencing method: Sequencing by synthesis

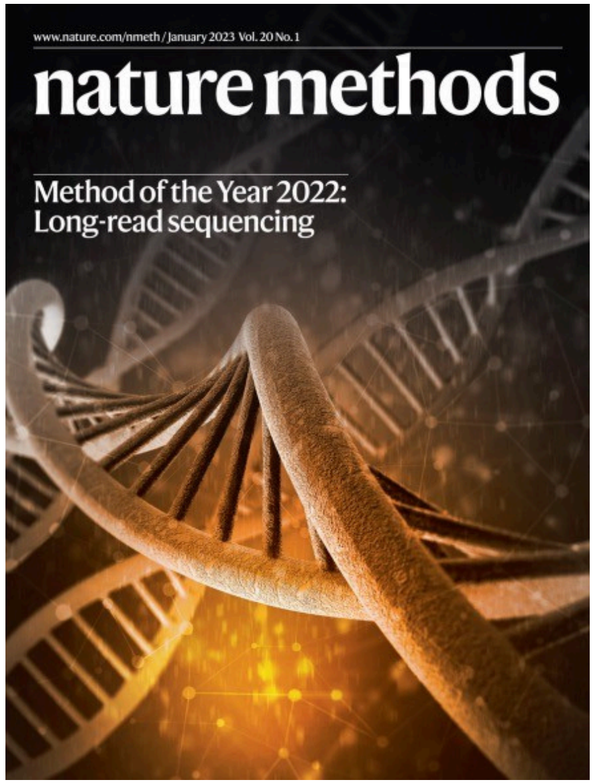



Tamang (2024)

# The Third Generation Sequencing Platforms: PacBio and Oxford Nanopore

**Lower throughput, but longer reads and additional information (base modifications)**



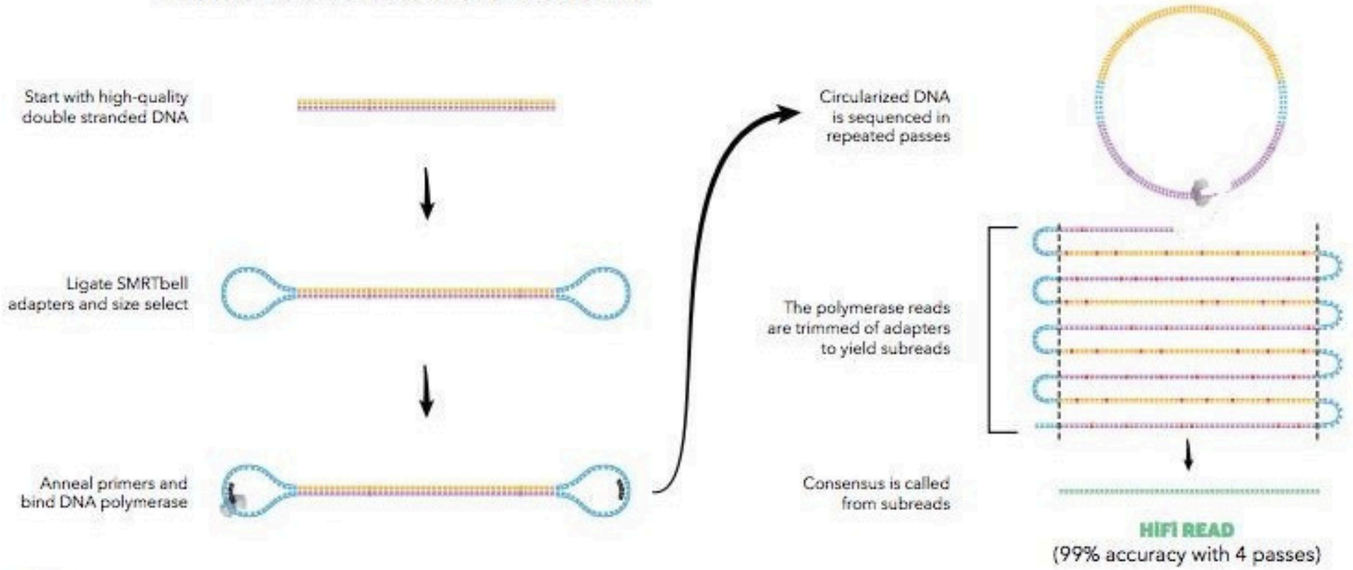



# HiFi Reads

HIFI READS

## Generate Highly Accurate Long Reads

Produce HiFi reads using the circular consensus sequencing (CCS) mode to provide base-level resolution for detection of all variant types from single nucleotide to structural variants.




LONG READS

## Optimize Your Run for Even Longer Reads

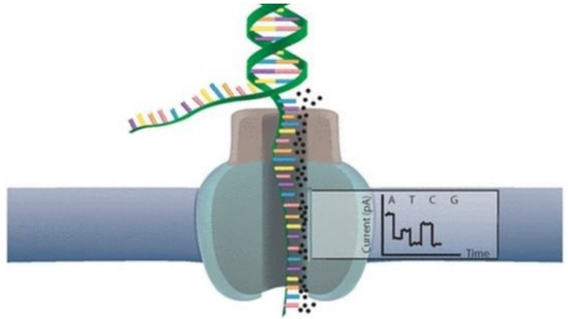
Sequence read lengths in the tens of kilobases using the continuous long read (CLR) sequencing mode to enable high-quality assembly of even the most complex genomes.

Half of Data in Reads
**>50 kb**

Longest Reads Up To
**175 kb**

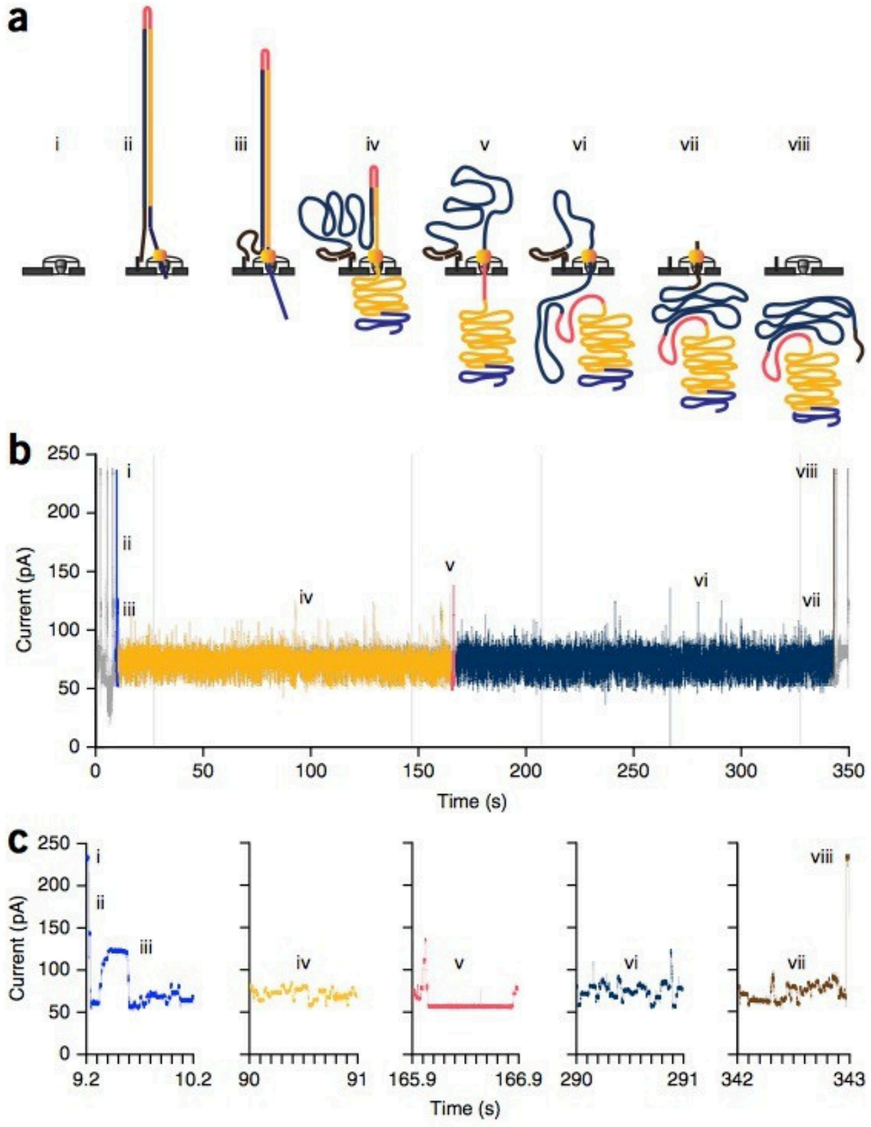
# Oxford Nanopore



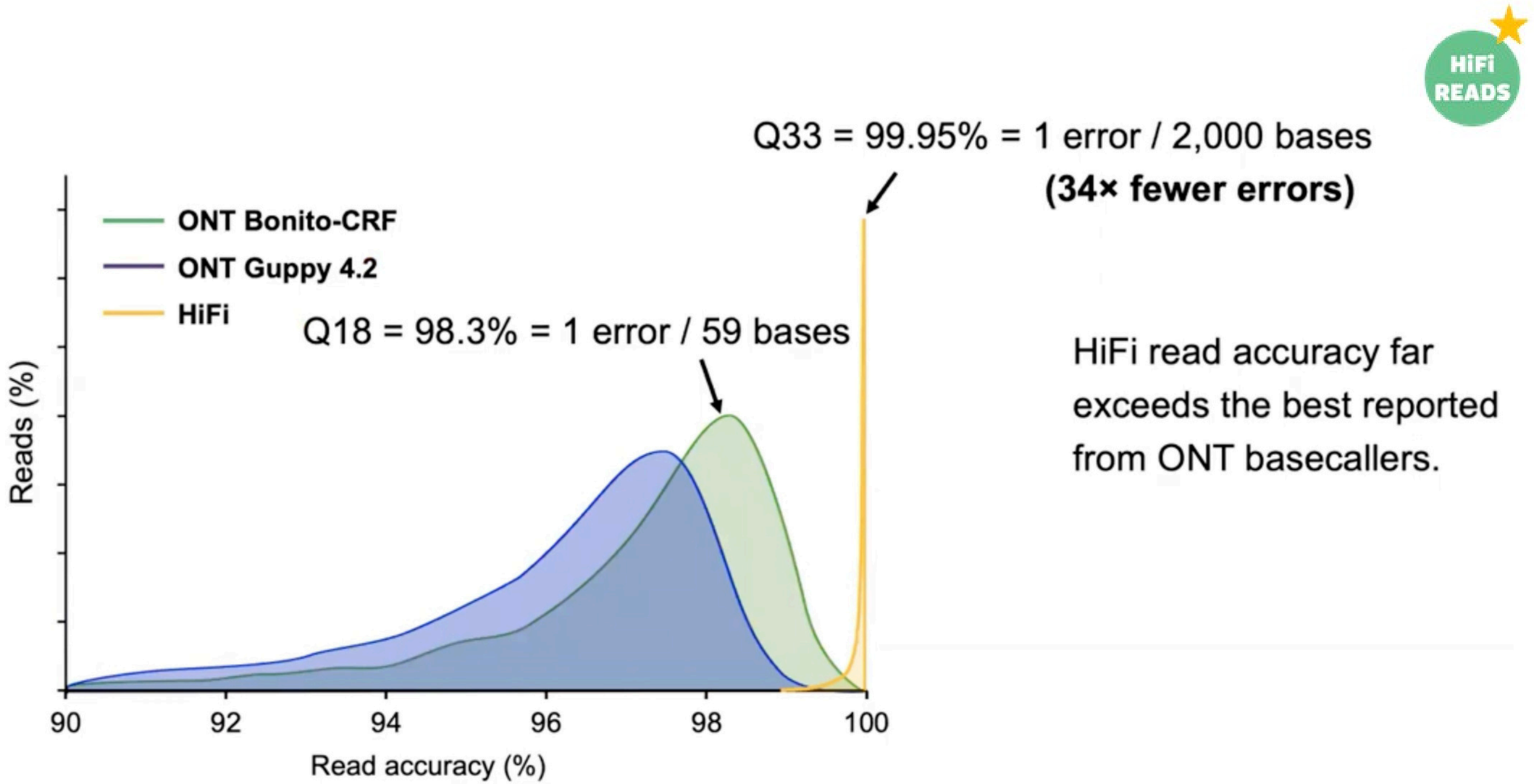




**Error rate still high!**
**indels 13%**
**substitution 5%**

**Recently improved**
**<2% with R10.4**



# HiFi reads compared to ONT




HiFi read accuracy far exceeds the best reported from ONT basecallers.

HiFi: HG003 18 kb library, Sequel II System Chemistry 2.0, precisionFDA Truth Challenge V2
ONT: Bonito-CRF & Guppy 4.2 NCM Nanopore Tech Update Dec. 2020

# Short reads vs Long reads

| Short reads | Long reads |
|---|---|
| **Illumina or MGI** | **PacBio (HiFi) or Nanopore** |
| Highly flexible usage | High quality genome assembly |
| | DNA base modifications detection |
| | |
| High quality reads of identical length | High quality HiFi reads (PacBio) |
| Enormous throughput, low price per base | Cheap apparatus (Nanopore) |
| **Limitations** | **Limitations** |
| Short (max 300bp PE) | Uneven length |
| Not suitable to solve repeats | High error rate in raw reads (mainly indels) |

# Advantages of long reads

High quality genome assembly

- Structural variation (such as inversions or insertions)
- Complex rearrangements
- Repeat expansion length variations
- Variants in repetitive or highly polymorphic regions
- Telomere-2-telomere
- Haplotigs separation or phasing

Other areas

- DNA base modifications detection
- Transcripts isoforms identification

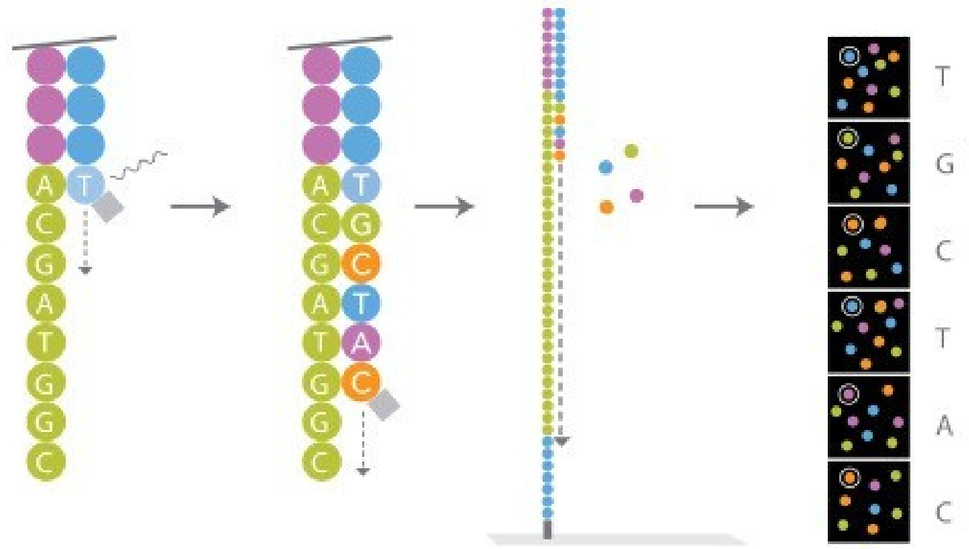
# Data cleaning & Quality control

# What do we receive from the sequencing facility?

The sequenced part of one DNA fragment.

These DNA sequences are given by the sequencing machine with a

Phred quality score in so called FASTQ format




```
@SRR001666.1  071112_SLXA-EAS1_s_7:5:1:817:345  length=36
TGCTACTGGCCGCTGCCGATGGCGTCAAATCCCACC
+SRR001666.1  071112_SLXA-EAS1_s_7:5:1:817:345  length=36
IIIIIIIIIIIIIIIIIIIIIIIIIIIII9IG9IC
```

# Phred quality score, a measure of base call quality




$$Q\text{sanger} = -10 \log_{10} p$$

Phred quality scores are logarithmically linked to error probabilities

| Phred Quality Score | Probability of incorrect call | Base call accuracy |
|---|---|---|
| 10 | 1 in 10 | 90% |
| 20 | 1 in 100 | 99% |
| 30 | 1 in 1000 | 99.9% |
| 40 | 1 in 10000 | 99.99% |
| 50 | 1 in 100000 | 99.999% |

The quality score is ASCII encoded in the FASTQ format

# FASTA vs FASTQ

```
>C3Po_0001:2:1:17:1499#0/1
TGAATTCATTGACCATAACAATCATATGCATGATGCAAATTATAATATCATTT
TTGTTTGAGCAAATGATTCATAATAATGTATTTCAATATTTTTAGGAATATCT
CCCAATATTGCGCGTGCTGAATTCCATCCGGAATTTTTGACGTCCCCCCCCGA
ANGGANGNGANNNNGNNGNNNNTNTNNAAANGNNNNN
…
```

```
@M01867:115:000000000-ABF5V:1:1101:9268:1666  1:N:0:51
AACAGGATTAGATACCCTGGTAGTCCACGCCCTAAACGATGCGAACTGGTTGTTGGGTGCTTTTTG
+
--A-6@8CE,@<CEFGGFAFF9CEFF,C@CE@B<8@C:CC,,+,7@C<6,668C,,+8,6,,<9,+
@M01867:115:000000000-ABF5V:1:1101:9214:1685  1:N:0:51
AACCGGATTAGATACCCTGGTAGTCCACGCCCTAAACGATGTCTACTAGTTGTTGGTGGAGTAAAA
+
--AA@7:FF9C9C@FEFE<CF9FEFF,C@FE:B8,6C:+C6CFD9CE,<C6<C@,,8,,,,;,,,-
@M01867:115:000000000-ABF5V:1:1101:18344:1708  1:N:0:51
AACAGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCAACTAGCCGTTGGGAGCCTTGAG
+
--A99E8CE<C9CFFGGG8FF9@CFF9ECFF@F;,CFC7C,CF,,CF@@EE@@@,,+,,6BE@,,-
…
```

read 1

read 2

read 3

FASTQ is a FASTA with score!

# Quality Control of the data

First step after receiving the data
Sometimes already done by the sequencing center

**Objective:**

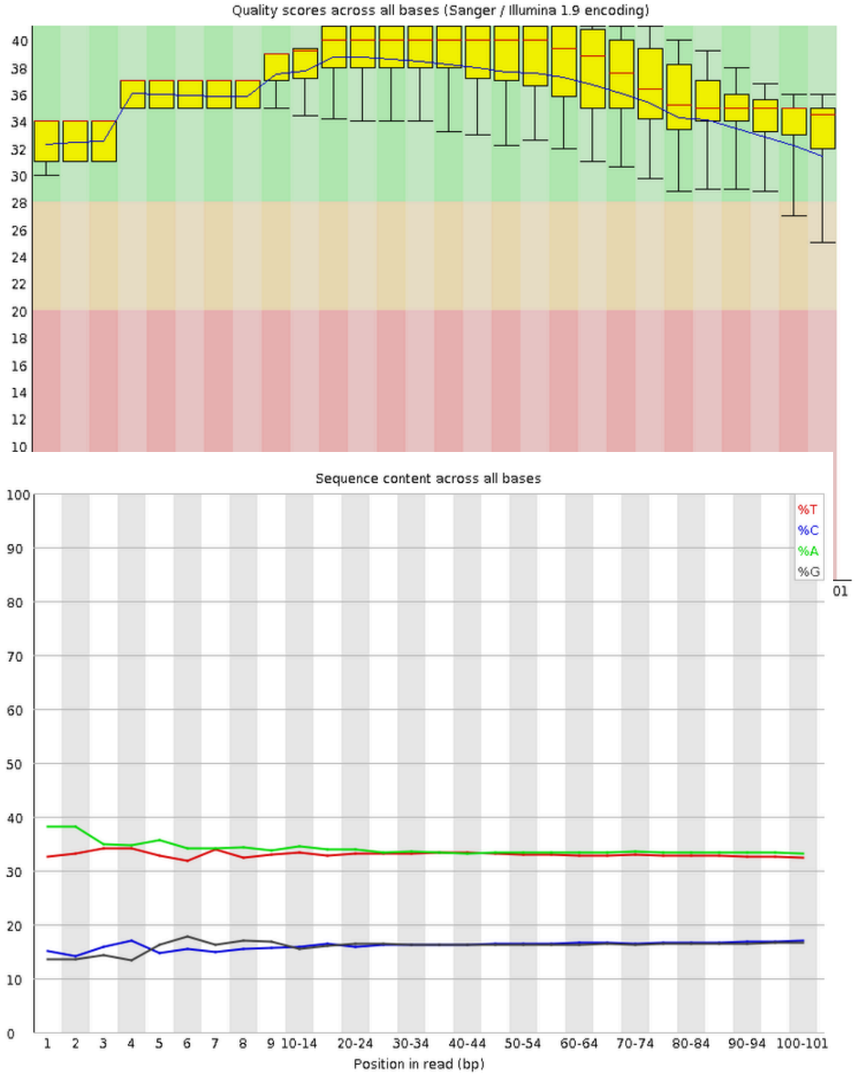
- Remove bad quality reads
- Remove contaminants Trim ends of reads
- Remove orphans (if possible or desirable)
- Correct errors

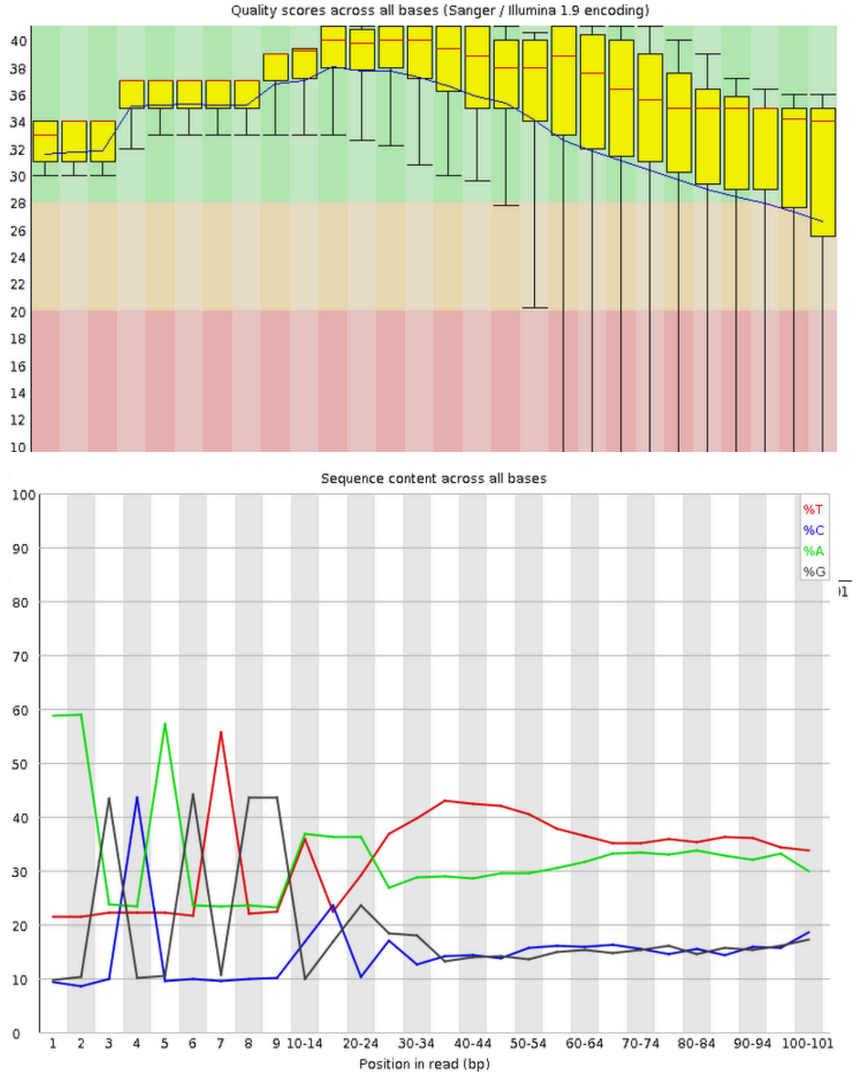



**Tools:**

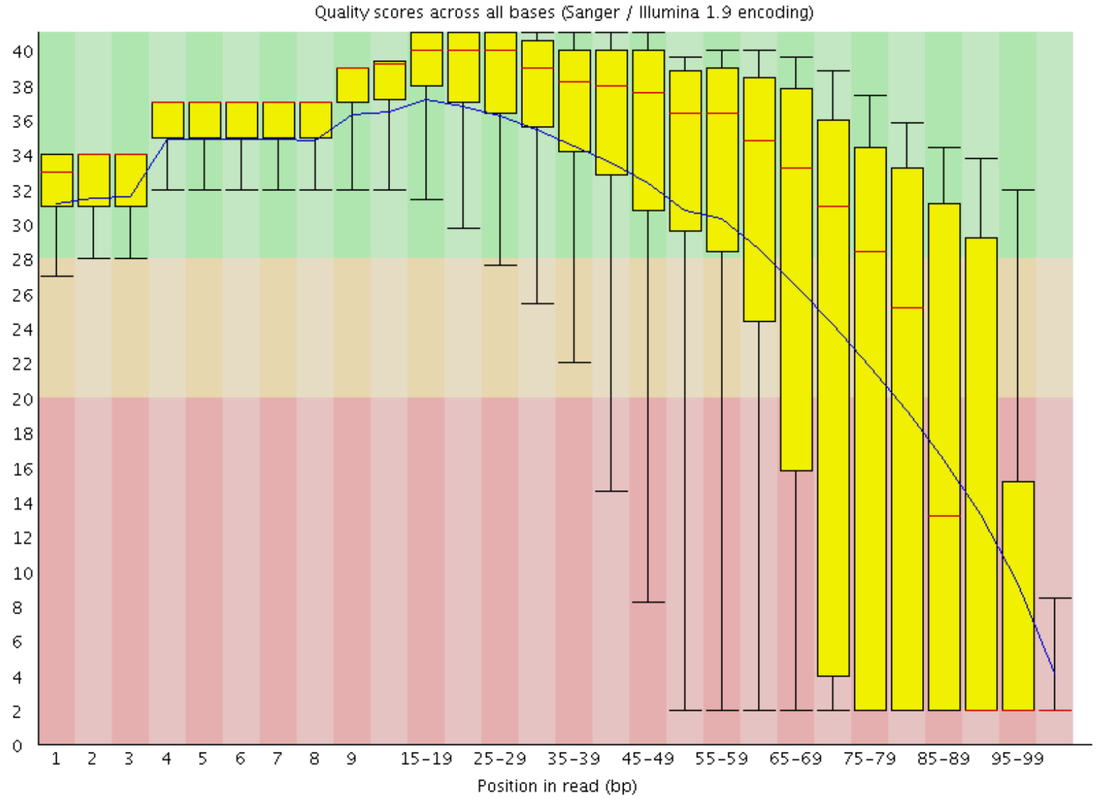
FastQC (http://www.bioinformatics.bbsrc.ac.uk/projects/fastqc/)
MultiQC (http://multiqc.info)
FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)

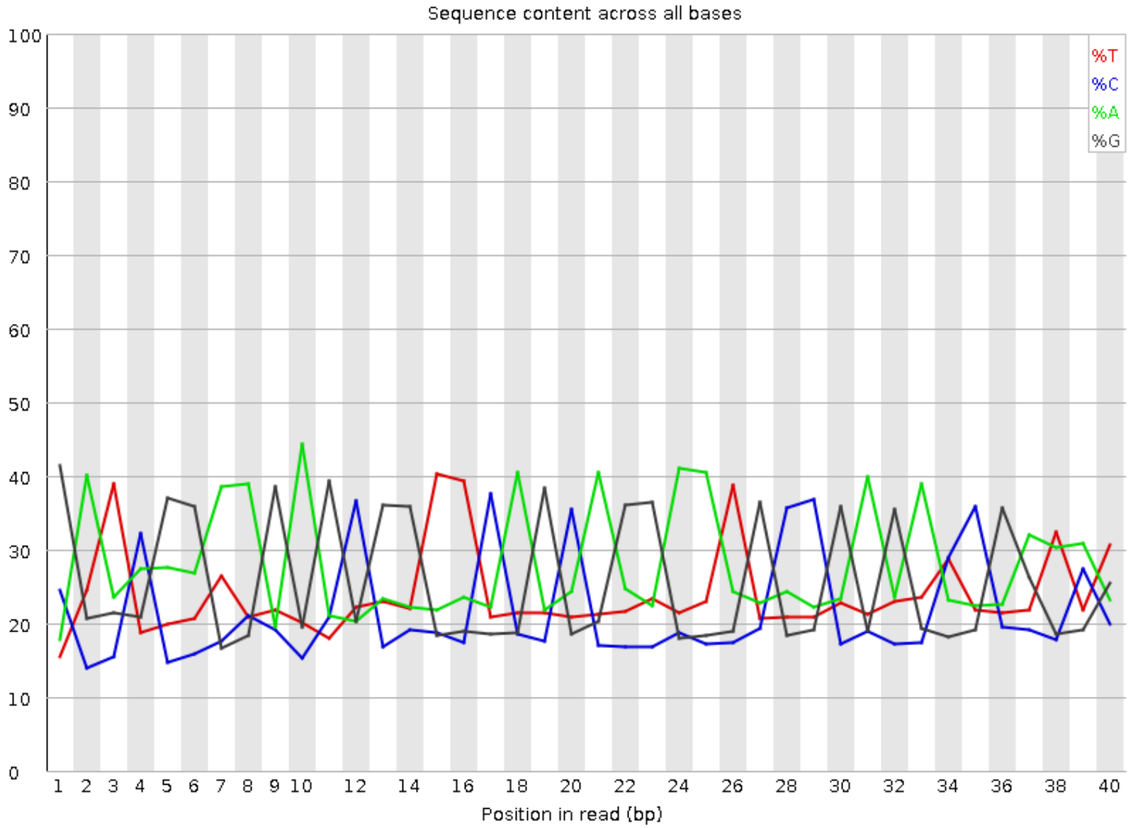
# Examples


Quality scores across all bases (Sanger / Illumina 1.9 encoding)
Sequence content across all bases
%T
%C
%A
%G
Position in read (bp)

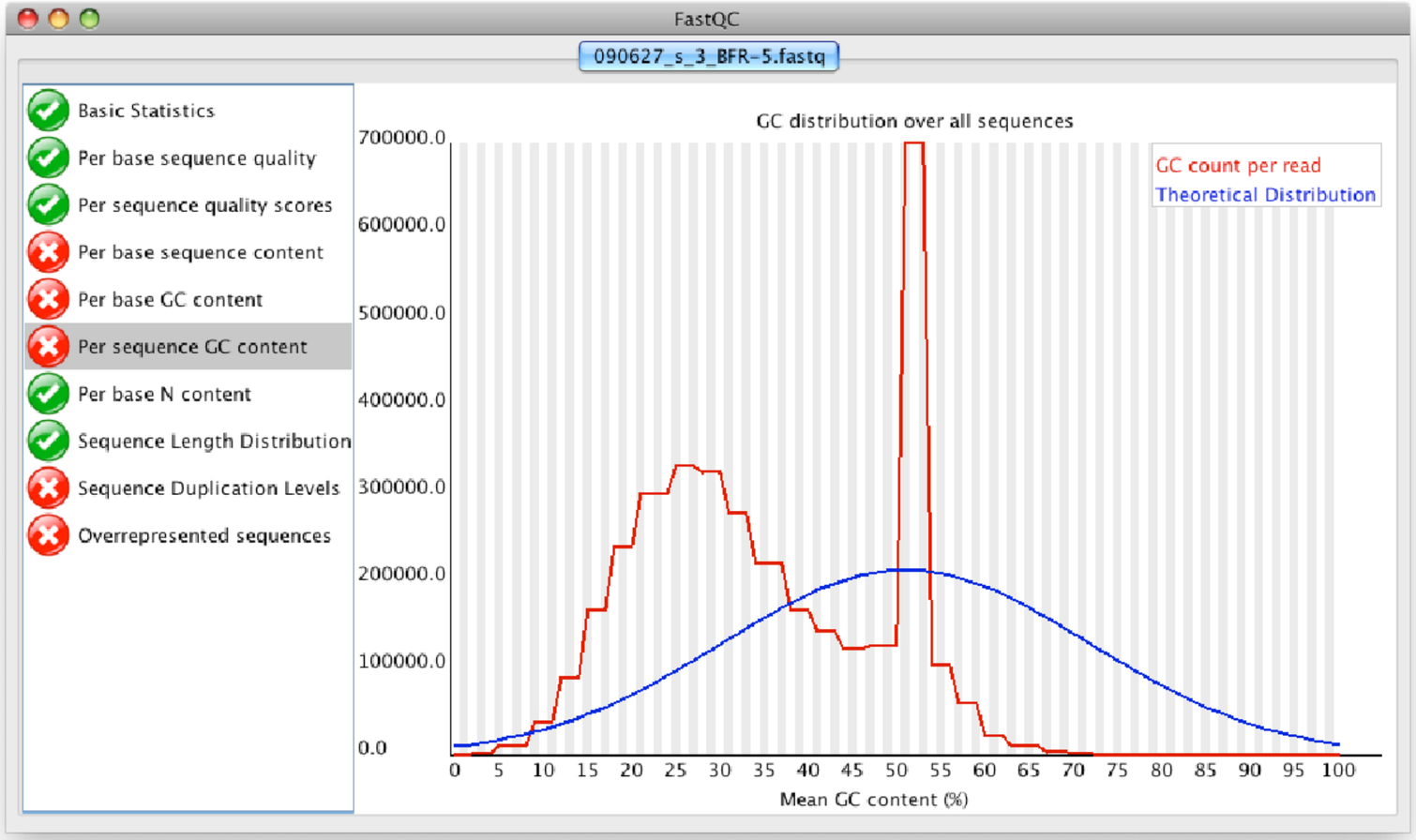

Quality scores across all bases (Sanger / Illumina 1.9 encoding)
Sequence content across all bases
%T
%C
%A
%G
Position in read (bp)

# Examples: good or bad?



# Examples: contamination



# Examples: contamination

## Overrepresented sequences

| Sequence | Count | Percentage | Possible Source |
|---|---|---|---|
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGATCT | 8122 | 8.122 | Illumina Paired End PCR Primer 2 (100% over 40bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGATCGGAAG | 5086 | 5.086 | Illumina Paired End PCR Primer 2 (97% over 36bp) |
| AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTAC | 1085 | 1.085 | Illumina Single End PCR Primer 1 (100% over 40bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGAAG | 508 | 0.508 | Illumina Paired End PCR Primer 2 (97% over 36bp) |
| AATTATACGGCGACCACCGAGATCTACACTCTTTCCCTAC | 242 | 0.242 | Illumina Single End PCR Primer 1 (97% over 40bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAAGATCGGAA | 235 | 0.23500000000000001 | Illumina Paired End Adapter 2 (96% over 31bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCGAGATCGGAAGA | 228 | 0.22799999999999998 | Illumina Paired End Adapter 2 (96% over 28bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACG | 205 | 0.20500000000000002 | Illumina Paired End PCR Primer 2 (97% over 36bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGATCGGAA | 183 | 0.183 | Illumina Paired End Adapter 2 (100% over 32bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGTCGGAAG | 183 | 0.183 | Illumina Paired End Adapter 2 (100% over 32bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGAACT | 164 | 0.164 | Illumina Paired End PCR Primer 2 (97% over 40bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGTCT | 129 | 0.129 | Illumina Paired End PCR Primer 2 (97% over 40bp) |
| AATTATACTTCTACCACCTATATCTACACTCTTTCCCTAC | 123 | 0.123 | No Hit |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGGACT | 122 | 0.122 | Illumina Paired End PCR Primer 2 (97% over 36bp) |
| CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGC | 113 | 0.11299999999999999 | Illumina Paired End PCR Primer 2 (96% over 25bp) |

# Examples: contamination

## Overrepresented sequences

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| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAAGATCGGAA | 235 | 0.23500000000000001 | Illumina Paired End Adapter 2 (96% over 31bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCGAGATCGGAAGA | 228 | 0.22799999999999998 | Illumina Paired End Adapter 2 (96% over 28bp) |
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| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGGTCGGAAG | 183 | 0.183 | Illumina Paired End Adapter 2 (100% over 32bp) |
| GATCGGAAGAGCGGTTCAGCAGGAATGCCGAGACCGAACT | 164 | 0.164 | Illumina Paired End PCR Primer 2 (97% over 40bp) |
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| CGGTTCAGCAGGAATGCCGAGATCGGAAGAGCGGTTCAGC | 113 | 0.11299999999999999 | Illumina Paired End PCR Primer 2 (96% over 25bp) |

# Other tools: MultiQC

tQC

is a quality control tool for high throughput sequence data, written by Simon Andrews at the Babraham Institute in Cambridge.

ıence Quality Histograms 7 4 1

ın quality value across each base position in the read. See the FastQC help.

Y-Limit

FastQC: Mean Quality Scores




Per Sequence GC Content 2 4 2

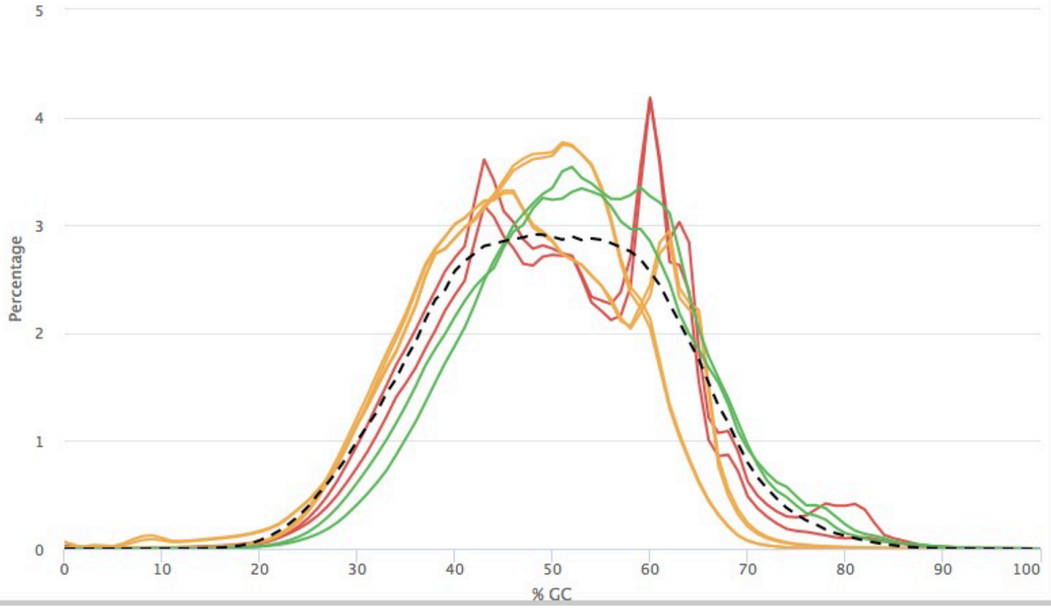
The dashed black line shows theoretical GC content: Human Transcriptome (UCSC hg38).

Y-Limits: on

Percentages Counts

FastQC: Per Sequence GC Content

Export Plot



# Read trimming or filtering

**Trimming** remove 5' and/or 3' ends of reads (bad quality or adapter)

**Filtering** remove full reads (e.g., contaminants)

**Tools:**

- FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
- PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)
- Sickle (https://github.com/najoshi/sickle)
- ea-utils (https://code.google.com/p/ea-utils/)
- Trimmomatic (http://www.usadellab.org/cms/?page=trimmomatic)
- cutadapt (https://cutadapt.readthedocs.org/)
- Fastp (https://github.com/OpenGene/fastp) and
- Fastplong (https://github.com/OpenGene/fastplong)

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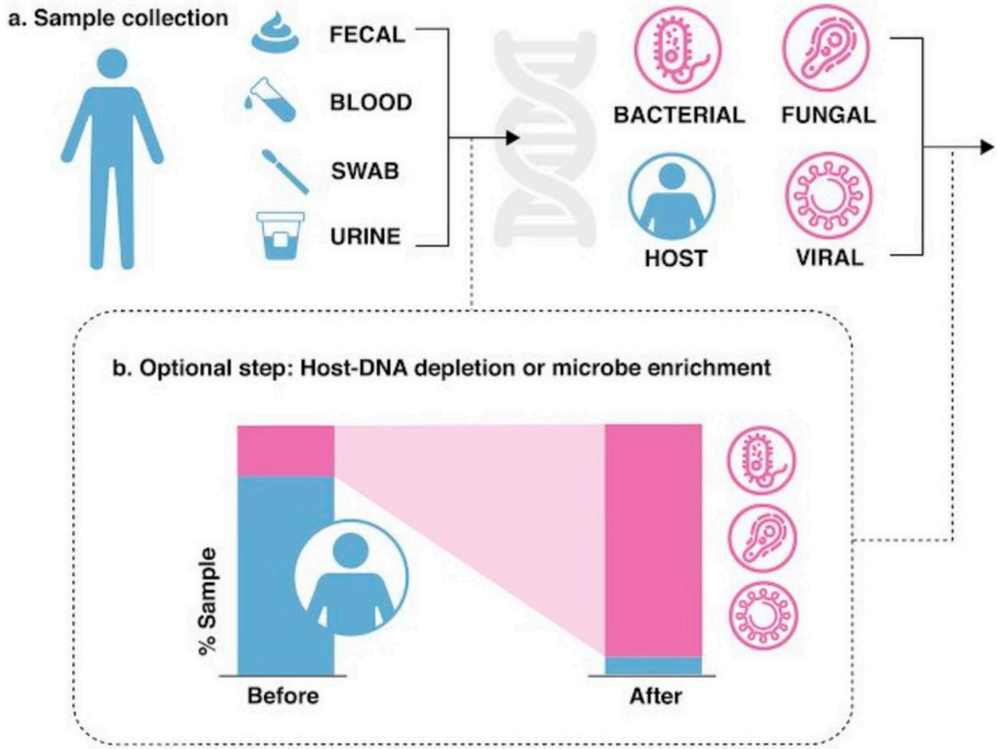
- FastX toolkit (http://hannonlab.cshl.edu/fastx_toolkit/)
- PrinSeq (http://edwards.sdsu.edu/cgi-bin/prinseq/prinseq.cgi)
- Sickle (https://github.com/najoshi/sickle)
- ea-utils (https://code.google.com/p/ea-utils/)
- Trimmomatic (http://www.usadellab.org/cms/?page=trimmomatic)
- cutadapt (https://cutadapt.readthedocs.org/)
- Fastp (https://github.com/OpenGene/fastp) and
- Fastplong (https://github.com/OpenGene/fastplong)

# Host removal or depletion

Host DNA sequences are very likely to contaminate your metagenomics data. It is critical to remove those sequences either before sequencing or before analysis.

E.g., PCR amplified 16S RNA can be contaminated by mitochondria or chloroplast 16S of the host.

E.g., WGS can be contaminated by host genomic DNA.



# Summary

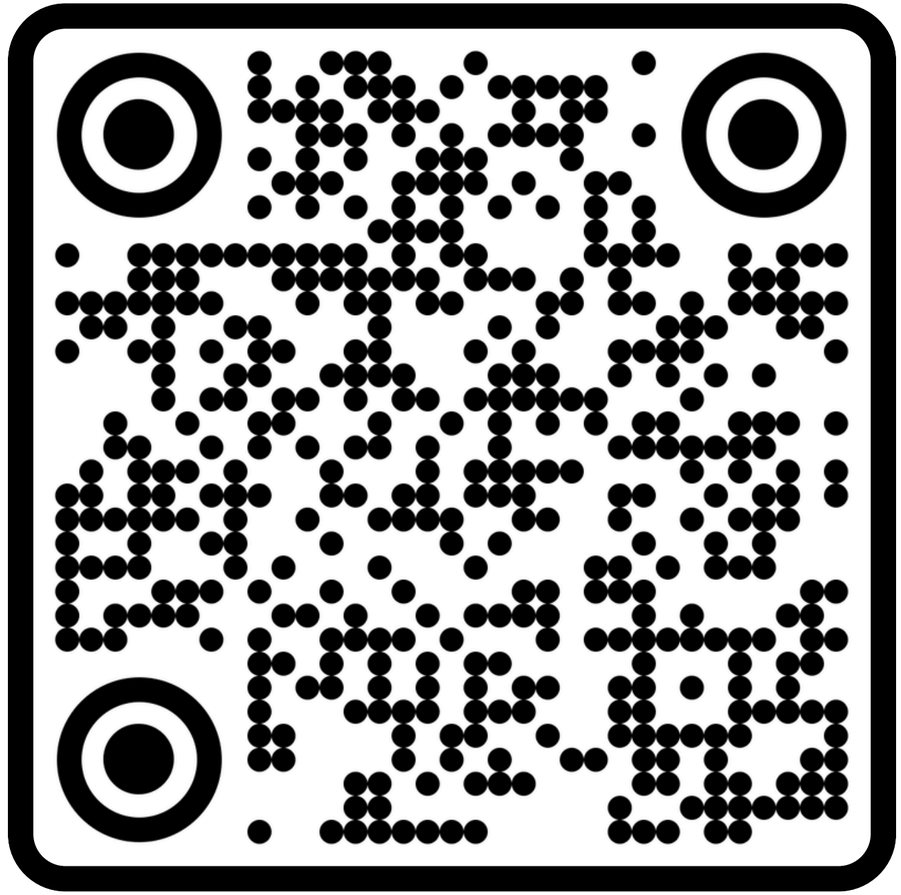
Different sequecing technologies are available

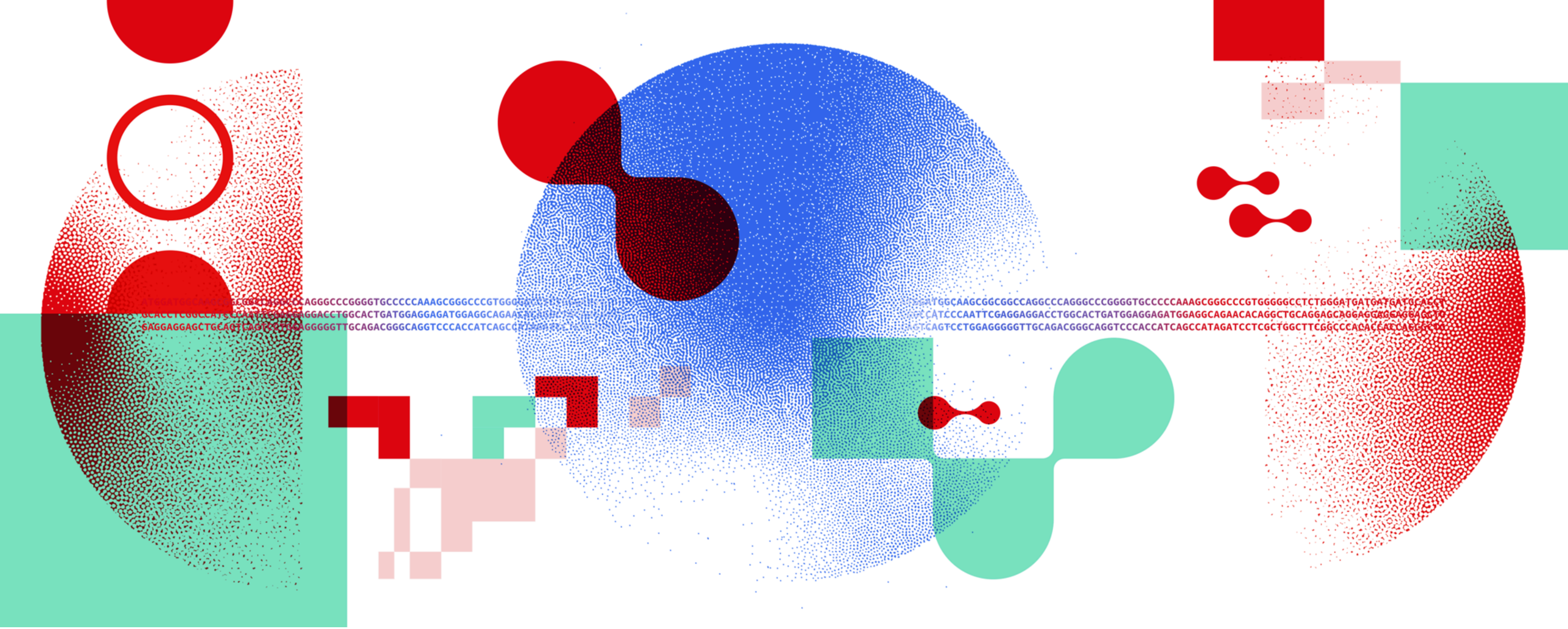
Sequecing technology should be chosen according to the goals of the study

Quality control is necessary to ensure reliability of the results

Analyze carefully and consciously the output of the quality control tools

# Practicals





# Thank you

elixir SWITZERLAND

SIB